Music Store Data Analysis

Presented by
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💰 Query 1: Customer Spending by Artist

Business Question:

How much did each customer spend on the BEST-SELLING ARTIST?

🎯 Business Impact:

Identify top customers for targeted marketingand loyalty programs

📊 Analyzing Customer Spending Patterns

This query uses a CTE (Common Table Expression) to:

Find the best-selling artist overall
Calculate how much EACH customer spent on that artist
Rank customers by their spending on this artist

💡 Skills Used:

- CTE (WITH clause)
- Multiple JOINs (6 tables)
- Aggregate Functions (SUM)
- GROUP BY & ORDER BY
- LIMIT for top artist

Customer spending on best-selling artist

```sql
WITH best_selling_artist AS (
    SELECT artist.artist_id AS artist_id, artist.name AS artist_name,
    SUM(invoice_line.unit_price * invoice_line.quantity) AS total_sales
    FROM invoice_line
    JOIN track ON track.track_id = invoice_line.track_id
    JOIN album ON album.album_id = track.album_id
    JOIN artist ON artist.artist_id = album.artist_id
    GROUP BY 1
    ORDER BY 3 DESC
    LIMIT 1
)

SELECT c.customer_id, c.first_name, c.last_name, bsa.artist_name,
SUM(il.unit_price * il.quantity) AS amount_spent
FROM invoice i
JOIN customer c ON c.customer_id = i.customer_id
JOIN invoice_line il ON il.invoice_id = i.invoice_id
JOIN track t ON t.track_id = il.track_id
JOIN album alb ON alb.album_id = t.album_id
JOIN best_selling_artist bsa ON bsa.artist_id = alb.artist_id
GROUP BY 1,2,3,4
ORDER BY 5 DESC;
```

🏆 TOP CUSTOMERS FOR BEST-SELLING ARTIST

Best-Selling Artist: QUEEN

Rank	Customer Name	Amount
1	Hugh O'Reilly	$27.72
2	Niklas Schroder	$18.81
3	Francois Tremblay	$17.82

📌 **Top spender spent 47% more than #2!**

💰 Query 2: Most Popular Genre by Country

Business Question:

What is the #1 music genre in EACH country?

🎯 Business Impact:

- Stock country-specific genres
- Targeted marketing campaigns
- Regional inventory optimization

📊 Genre Analysis Across Countries

This query uses a WINDOW FUNCTION to:

Count purchases by genre in each country
Rank genres within each country
Return only the #1 genre per country

💡 Advanced SQL Skills:

- Window Function (ROW_NUMBER)
- PARTITION BY country
- CTE (WITH clause)
- Multiple JOINs (5 tables)

Most popular genre in each country

```sql
WITH popular_genre AS (
    SELECT COUNT(invoice_line.quantity) AS purchases, customer.country, genre.name, genre.genre_id,
    ROW_NUMBER() OVER(PARTITION BY customer.country ORDER BY COUNT(invoice_line.quantity) DESC) AS RowNo
    FROM invoice_line
    JOIN invoice ON invoice.invoice_id = invoice_line.invoice_id
    Join customer ON customer.customer_id = invoice.customer_id
    JOIN track ON track.track_id = invoice_line.track_id
    JOIN genre ON genre.genre_id = track.genre_id
    GROUP BY 2,3,4
    ORDER BY 2 ASC, 1 DESC
)
SELECT * FROM popular_genre WHERE ROWNO <= 1
```

🌍 TOP GENRE IN EACH COUNTRY

Top Genre: Rock

Country	Top Genre
Argentina	Alternative & Punk
Australia	Rock
Austria	Rock
Belgium	Rock

📌 **Insight: Rock dominates in 60% of countries!**

💰 Query 3: Top Customer in Each Country

Business Question:

Who is the highest-spending customer in EACH country?

🎯 Business Impact:

- Identify VIP customers by region
- Country-specific loyalty programs
- Targeted promotions for top spenders

📊 Advanced Analysis with Recursive CTE

This query uses a RECURSIVE CTE to:

Calculate total spending per customer by country
Find maximum spending in each country
Match customers who achieved that maximum

💡 Advanced SQL Skills:

- RECURSIVE CTE (rare and powerful!)
- Multiple CTEs in one query
- JOIN between CTEs
- Window-level analysis

Top customer in each country

```sql
WITH RECURSIVE
    customer_with_country AS (
        SELECT customer.customer_id, first_name, last_name, billing_country, SUM(total) AS total_spending
        FROM invoice
        JOIN customer ON customer.customer_id = invoice.customer_id
        GROUP BY 1,2,3,4
        ORDER BY 1,5 DESC
    ),

    country_max_spending AS(
        SELECT billing_country, MAX(total_spending) AS max_spending
        FROM customer_with_country
        GROUP BY billing_country
    )

SELECT cc.billing_country, cc.total_spending, cc.first_name, cc.last_name
FROM customer_with_country cc
JOIN country_max_spending ms
ON cc.billing_country = ms.billing_country
WHERE cc.total_spending = ms.max_spending
ORDER BY 1
```

👑 TOP CUSTOMER BY COUNTRY

R Madhav (Czech Republic) - $144.54

Country	Top Customer	Total Spent
Argentina	Diego Gutierrez	$39.60
Belgium	Daan Peeters	$60.39
Czech Republic	R Madhav	$144.54
Brazil	Luís Gonçalves	$108.90

🏆 **Top spender: R Madhav (Czech Republic) - $144.54**

Contact Us

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GitHub

https://github.com/afaqahmad2005/Music-Store-Data-Analysis-Project-using-SQL